GPU-Accelerated Density Functional Theory Solver for Binary SALR Colloids (SALR3YUCUDA)

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***Abstract*—We present a computational framework for solving mean-field density functional theory (DFT) equations describing spatial density distributions of binary colloidal mixtures with competing short-range attraction and long-range repulsion (SALR) interactions. The solver implements the Picard iterative method for a two-component system with different walls configuration, using the triple Yukawa potential model. Our CPU implementation employs OpenMP parallelization, achieving significant speedup on multi-core processors. A robust CUDA implementation for GPUs was developed, delivering up to $\approx 40\times$ faster runtime compared to the single-threaded CPU version and a $\sim 5\times$ speedup compared to maximum OpenMP parallelization. We demonstrate the solver capabilities by computing equilibrium density profiles under periodic, two-wall, and four-wall boundary conditions. A Qt-based graphical user interface (GUI) was also built to facilitate parameter configuration, task execution, and result visualization.**

I. Introduction

Colloidal systems provide important models for studying fundamental phenomena in condensed matter physics, including phase transitions, self-assembly, and complex fluid behavior [**?**]. Among the most intriguing colloidal systems are those exhibiting competing interactions, where particles experience short-range attraction and long-range repulsion (SALR). These interactions, sometimes called "mermaid" potentials due to their attractive head and repulsive tail [**?**], lead to rich phase behavior including microphase separation, cluster formation, and periodic stripe patterns [**?**]. Figure **??** illustrates a typical SALR potential profile, with the short-range attractive well ("head") followed by a long-range repulsive barrier ("tail").

Despite significant theoretical predictions about SALR systems, experimental realization of the predicted mesophases in three-dimensional real space remains challenging. As Royall notes, "no ordered phase in a system with competing interactions has ever been observed in 3d real space" [**?**]. This gap between theory and experiment motivates the development of efficient computational tools that allow researchers to explore parameter space rapidly and identify conditions under which specific morphologies may emerge.

The theoretical description of spatial density distributions in SALR systems is commonly performed using density functional theory (DFT) within the mean-field approximation [**?**]. This approach leads to a system of nonlinear integral equations that must be solved iteratively. For a two-dimensional grid of $N_x \times N_y$ nodes, the computational cost per iteration scales as $O(N^2)$ due to the convolution required to compute interaction fields, making the problem computationally intensive.

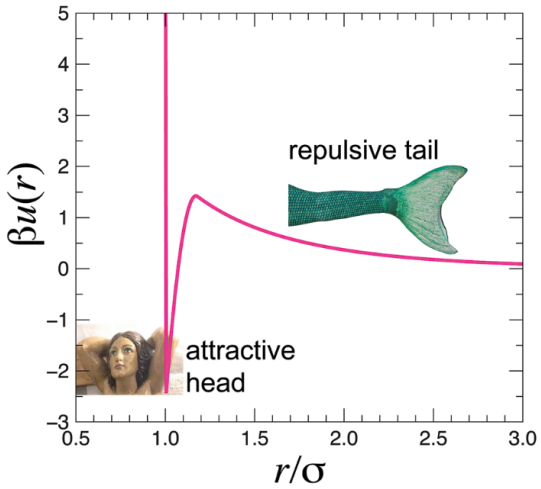

Fig. 1. A "mermaid" (SALR) pair potential as a function of reduced separation r/σ. The attractive head promotes local condensation while the repulsive tail prevents bulk phase separation, driving the system toward microphase-separated morphologies such as clusters and stripes. Reproduced from [**?**].

In this work, we present SALR3YUCUDA - a solver for mean-field DFT equations describing binary SALR mixtures. The project name reflects its key components: SALR (Short-range Attraction Long-range Repulsion), 3Y (triple Yukawa potential), and CUDA (GPU acceleration technology). We implemented an optimized pure C version with OpenMP parallelization for scalable execution on multi-core processors, accompanied by a highly parallelized CUDA framework for GPU-accelerated massive parameter computations. The solvers natively integrate with an SQLite-HDF5 database for session tracking and large spatial tensor storage, and feature multiple boundary conditions including periodic boundaries (PBC), two parallel walls (W2), and four-wall confinement (W4). Furthermore, a Qt-based graphical interface enables rich visual exploration without cumbersome command-line workflows.

A. Motivation

Binary colloidal mixtures with SALR interactions are of fundamental interest across soft matter, biophysics, and materials science. The interplay of competing attraction and repulsion gives rise to self-assembled mesophases—clusters, stripes, and lamellar structures—that serve as model systems for understanding protein aggregation, lipid membranes, and block-copolymer microphase separation [**?**], [**?**]. Mapping the conditions under which each morphology appears requires systematic exploration of a high-dimensional parameter space (composition, temperature, interaction strengths and ranges), which is experimentally costly and time-consuming.

Particle-resolved computer simulations such as molecular dynamics (MD) and Monte Carlo (MC) provide the most detailed description of these systems, tracking every particle trajectory and configuration. However, their computational cost scales with the number of particles N_p: a single equilibration run for a system large enough to contain several stripe periods may require millions of MC sweeps or nanoseconds of MD trajectory, making large-scale parameter surveys impractical. Furthermore, finite-size effects are pronounced when the system box is not much larger than the characteristic pattern wavelength.

Density functional theory offers a complementary approach that avoids these limitations. Rather than evolving individual particle positions, DFT works directly with continuous density fields $\rho_i(\mathbf{r})$, reducing an N_p-body statistical mechanics problem to a deterministic nonlinear integral equation solved on an $N_x \times N_y$ grid. Key advantages include:

- **Speed:** A single DFT iteration costs $O(N_x^2 N_y^2)$ operations, but convergence typically requires only 10^2–10^4 Picard steps, compared to 10^6–10^8 MC sweeps for particle simulations of equivalent domain size.
- **Direct equilibrium access:** The iterative solver finds the equilibrium density profile directly without the need for thermalization or ensemble averaging inherent to stochastic simulations.
- **Clean boundary conditions:** Walled geometries are imposed analytically as Dirichlet constraints on ρ_i, avoiding surface-energy artifacts from finite particle layers.
- **Parameter scalability:** Changing temperature, density, or interaction coefficients requires only reinitializing the solver, enabling rapid sweeps across the phase diagram.

These properties make mean-field DFT the method of choice for initial reconnaissance of SALR phase behavior, providing guidance for more expensive particle-level studies and for comparison with theory. The present work implements and accelerates this approach, with GPU parallelization targeting the dominant $O(N^2)$ convolution kernel to rapidly expand the accessible parameter space.

II. Theoretical Background

A. SALR Interactions

Short-range attraction and long-range repulsion (SALR) interactions arise naturally in colloidal systems where particles experience depletion attraction at short distances (due to polymer or small particle exclusion) and electrostatic repulsion at longer distances [**?**]. The competition between these interactions leads to complex energy landscapes and the possibility of microphase separation, where the system forms periodic structures such as clusters, stripes, or lamellae rather than undergoing bulk phase separation.

B. Triple Yukawa Potential

We model the pair interaction between particles of species i and j using a triple Yukawa form:

$$U_{ij}(r) = \sum_{m=1}^{3} A_{ij}^{(m)} \frac{\exp(-\alpha_{ij}^{(m)} r)}{r}, \quad 0 < r \le r_c \tag{1}$$

where $A_{ij}^{(m)}$ are amplitude coefficients, $\alpha_{ij}^{(m)}$ are inverse screening lengths, and r_c is a cutoff radius beyond which $U_{ij}(r) = 0$. The three Yukawa terms allow flexible representation of the SALR shape: positive amplitudes at short inverse lengths create repulsion, while negative amplitudes create attraction.

For a binary mixture (species 1 and 2), we require three independent potentials: $U_{11}(r)$ for species 1–1 interactions, $U_{22}(r)$ for species 2–2 interactions, and $U_{12}(r) = U_{21}(r)$ for cross-species interactions.

C. Mean-Field Density Functional Theory

Within the mean-field approximation, the grand thermodynamic potential functional for the binary mixture leads to Euler-Lagrange equations that determine the equilibrium density profiles $\rho_1(\mathbf{r})$ and $\rho_2(\mathbf{r})$:

$$\rho_1(\mathbf{r}) = \rho_{1,b} \exp[-\beta\,(\Phi_{11}(\mathbf{r}) + \Phi_{12}(\mathbf{r}) - \Phi_{11,b} - \Phi_{12,b})] \tag{2}$$

$$\rho_2(\mathbf{r}) = \rho_{2,b} \exp[-\beta\,(\Phi_{21}(\mathbf{r}) + \Phi_{22}(\mathbf{r}) - \Phi_{21,b} - \Phi_{22,b})] \tag{3}$$

where $\beta = 1/(k_B T)$ is the inverse temperature (with k_B the Boltzmann constant), $\rho_{i,b}$ is the bulk (average) density of species i, and $\Phi_{ij}(\mathbf{r})$ are interaction field contributions defined as convolutions:

$$\Phi_{ij}(\mathbf{r}) = \int d\mathbf{r}'\, \rho_j(\mathbf{r}')\, U_{ij}(|\mathbf{r} - \mathbf{r}'|) \tag{4}$$

The bulk field values $\Phi_{ij,b}$ represent the interaction fields in a uniform system with densities $\rho_{1,b}$ and $\rho_{2,b}$, ensuring that the trivial homogeneous solution satisfies the equations.

D. Picard Iteration

Equations (**??**)–(**??**) are solved iteratively using the Picard method with mixing:

$$\rho_i^{(t+1)}(\mathbf{r}) = \xi_i K_i[\rho^{(t)}](\mathbf{r}) + (1 - \xi_i)\rho_i^{(t)}(\mathbf{r}) \tag{5}$$

where $K_i[\rho]$ denotes the right-hand side of the Euler-Lagrange equation and $\xi_i \in (0, 1]$ is a mixing parameter that controls convergence stability. Iterations continue until the root-mean-square change falls below a tolerance ε:

$$\left\| \rho_i^{(t+1)} - \rho_i^{(t)} \right\| = \sqrt{\frac{1}{N_x N_y} \sum_k \left[{\rho_i^{(t+1)}}_k - {\rho_i^{(t)}}_k \right]^2} < \varepsilon \tag{6}$$

E. Boundary Conditions

The solver supports three boundary modes:

- **PBC (Periodic Boundary Conditions):** The domain wraps around in both x and y directions, simulating a bulk-like infinite system.
- **W2 (Two Walls):** Hard walls at $x = 0$ and $x = L_x$ with periodic boundaries in y. The density is forced to zero at wall positions: $\rho_i(x = 0) = \rho_i(x = L_x) = 0$.
- **W4 (Four Walls):** Hard walls on all boundaries, confining particles to a rectangular box.

III. Implementation

A. Discretization

The two-dimensional domain $[0, L_x] \times [0, L_y]$ is discretized on a uniform grid with $N_x \times N_y$ nodes and spacing Δx, Δy. The continuous convolution integral (**??**) becomes a discrete sum:

$$\Phi_{ij}(x_i, y_j) = \Delta x \Delta y \sum_{m,n} \rho_j(x'_m, y'_n)\, U_{ij}(|\mathbf{r}_{ij} - \mathbf{r}'_{mn}|) \quad (7)$$

For efficiency, the potential $U_{ij}(r)$ is precomputed and stored in lookup tables indexed by displacement (d_{ix}, d_{iy}) between grid cells.

B. Algorithm Overview

The solver proceeds through the following steps:

1) **Initialization:** Load parameters; build potential tables U_{11}, U_{12}, U_{22}; compute bulk fields $\Phi_{ij,b}$; set initial density profiles with small perturbations.
2) **Iteration loop:**
   a) Compute interaction fields Φ_{11}, Φ_{12}, Φ_{21}, Φ_{22} via discrete convolution.
   b) Evaluate K_1 and K_2 using equations (**??**)–(**??**).
   c) Apply mass conservation renormalization.
   d) Perform Picard mixing (**??**).
   e) Apply boundary conditions (zero density at walls).
   f) Apply anti-aliasing smoothing to suppress numerical artifacts.
   g) Check convergence criterion (**??**).
3) **Output:** Save converged density profiles.

C. CPU Parallelization

The most computationally expensive operation is the $O(N^2)$ convolution for computing Φ_{ij}. We employ OpenMP to parallelize:

- Potential table construction (loop over grid offsets)
- Convolution computation (nested loops over source and field cells)
- K-operator evaluation and Picard mixing

On a multi-core processor (Intel Core i7-13700H, 20 threads), this OpenMP parallelization achieves up to a 7× speedup compared to single-threaded execution on the largest grid sizes (150×150).

D. CUDA Implementation

GPU acceleration explicitly targets the critical convolution operation by offloading both potential generation and density mixing to the CUDA kernel pipeline. We employ coalesced global memory access and constant memory for potential parameters (such as the Yukawa amplitudes and inverse screening lengths) to maximize global bandwidth throughput. Custom 2D thread blocks iteratively reduce memory fetches during the spatial convolution loops.

E. GUI and Session Database

The project integrates a comprehensive Qt 6-based Graphical User Interface developed for end-to-end simulation lifecycle management. Parameters such as temperature, grid shape, and boundary constraints are defined seamlessly. More essentially, the GUI wraps an underlying SQLite and HDF5 backend which ensures persistent tracking of thermodynamic constraints, density tensors over iterative runs, and allows live visual representation (heatmaps and scatter plots) directly from the persistent '.h5' artifacts, eliminating the overhead of manual data handling.

Fig. 2. Screenshot of the Qt Visualization GUI. The interface features parameter control, run initialization, database session browsing, and real-time visualization of density fields via heatmaps and scatter plots.

IV. Results

A. Simulation Parameters

We demonstrate the solver using the parameters listed in Table **??**. The temperature $T = 2.9$ places the system in a regime where microphase separation occurs, producing visible density modulations.

The triple Yukawa coefficients for species 1–1 interaction are $A_{11}^{(1)} = 150.66$, $A_{11}^{(2)} = -122.61$, $A_{11}^{(3)} = 27.12$ with

TABLE I
SIMULATION PARAMETERS

Parameter	Value
System size $L_x \times L_y$	16.0×16.0
Grid resolution $N_x \times N_y$	80×80
Grid spacing $\Delta x = \Delta y$	0.2
Temperature T	2.9
Bulk density $\rho_{1,b}$	0.4
Bulk density $\rho_{2,b}$	0.2
Cutoff radius r_c	8.0
Mixing coefficients $\xi_1 = \xi_2$	0.1
Tolerance ε	10^{-8}

corresponding $\alpha_{11}^{(1)} = 1.92$, $\alpha_{11}^{(2)} = 1.26$, $\alpha_{11}^{(3)} = 0.76$. These parameters produce a potential with attractive head and repulsive tail characteristic of SALR systems.

B. Periodic Boundary Conditions (PBC)

Figure **??** shows the converged density distribution under periodic boundary conditions. The system exhibits vertical stripe formation, with alternating regions of high species-1 density (purple) and high species-2 density (green). This demixing pattern reflects the competition between short-range attraction (favoring condensation) and long-range repulsion (preventing bulk separation).

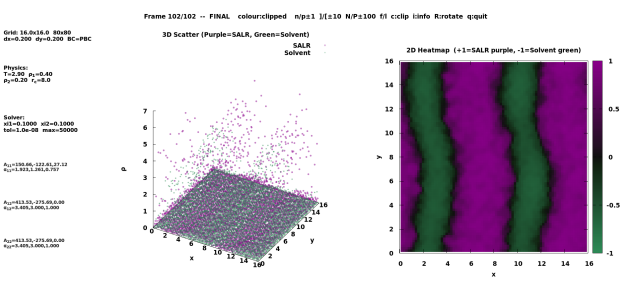

Fig. 3. Density distribution under periodic boundary conditions (PBC). Left: 3D scatter plot showing species 1 (purple) and species 2 (green). Right: 2D heatmap overlay. Vertical stripes indicate microphase separation with characteristic wavelength determined by competing interaction scales.

C. Two-Wall Confinement (W2)

Figure **??** presents results for confinement between two parallel walls at $x = 0$ and $x = L_x$. The hard-wall boundary condition ($\rho = 0$ at walls) fundamentally alters the density distribution compared to PBC:

- Density depletes near the walls, creating boundary layers where both species have reduced concentration.
- Stripes orient parallel to the walls (perpendicular to the confinement direction), as this orientation minimizes the free energy penalty from wall interactions.
- The periodic direction (y) allows stripe continuity, producing a layered structure.

D. Four-Wall Confinement (W4)

Figure **??** shows the fully confined case with hard walls on all four boundaries. The box geometry produces:

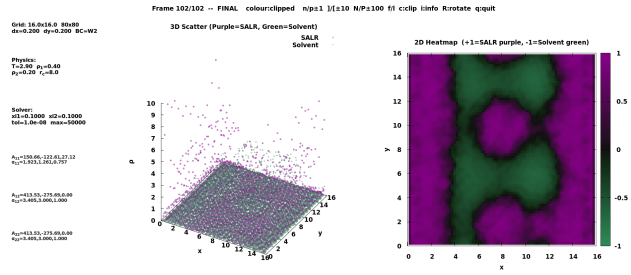

Fig. 4. Density distribution with two-wall confinement (W2). Hard walls at $x = 0$ and $x = L_x$ induce depletion layers and reorient stripes parallel to the confining surfaces.

- A depleted boundary layer surrounding the entire perimeter.
- Complex internal structure with density concentrated in the central region.
- Corner effects where the two perpendicular walls create regions of severe depletion.
- Species segregation within the confined volume, with the higher-density species (species 1) forming a more compact central distribution.

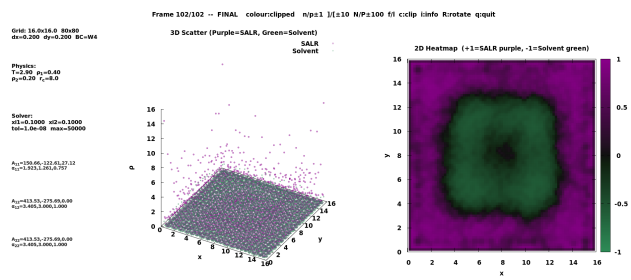

Fig. 5. Density distribution with four-wall confinement (W4). Complete enclosure produces boundary depletion on all edges and concentrated density in the central region.

E. Performance

A scalability analysis details the runtime throughput in terms of convolution evaluation per grid volume. Using a 150×150 grid, the single-threaded OpenMP approach required 764s execution time. Under maximal (20-thread) CPU allocation, time converged bounding closer to 100s.

In sharp contrast, the corresponding execution on an NVIDIA GeForce RTX 4060 Laptop GPU achieved complete iteration to convergence in just 19.2s. This corresponds to an aggregated speedup over $39.7\times$ against the fundamental serial baseline and a $\sim 5\times$ speed gain explicitly over full CPU thread saturation. Figure **??** demonstrates the drastic acceleration achieved via the CUDA framework against variable OpenMP implementations.

V. DISCUSSION

The computed density distributions demonstrate that our solver correctly captures the essential physics of SALR sys-

Fig. 6. Overall Speedup (vs. single-threaded CPU runtime) corresponding to variable problem geometries (16×16 to 150×150) emphasizing the asymptotic supremacy of execution on CUDA architecture.

tems: microphase separation driven by competing interactions, and the influence of confinement on pattern morphology.

The observation of vertical stripes under PBC and wall-parallel stripes under W2 confinement aligns with theoretical expectations. In the bulk-like PBC case, symmetry is broken by the initial conditions, selecting a particular stripe orientation. When walls are introduced, the system minimizes its free energy by aligning stripes parallel to the confining surfaces, reducing the density gradient penalty at the boundaries.

These results support the utility of computational tools for exploring parameter space in SALR systems. As noted by Royall [**?**], experimental realization of predicted mesophases faces challenges including weak ion dissociation in non-aqueous solvents and non-centrosymmetric interactions. Simulations in idealized conditions allow researchers to establish baseline expectations before confronting experimental complexities.

Specifically, the deployment on CUDA effectively shrinks days-long iterative experiments on large grids to mere minutes, bringing rapid exploration and phase-space sweeping into realistic ranges. While CPU cores exhibit efficiency plateaus and bounded bandwidths as cache lines overextend during grid sweeps, the CUDA architecture inherently processes continuous convolutions effectively, maintaining almost steady-state high throughput unimpacted by sequential bounds.

VI. Conclusion

We have presented SALR3YUCUDA, a computational solver for mean-field DFT equations describing binary SALR colloidal mixtures. The CPU implementation with OpenMP parallelization and, crucially, a highly optimized GPU accelerator in CUDA successfully compute equilibrium density distributions under various boundary conditions, demonstrating stripe and cluster formation characteristic of competing-interaction systems in fractions of previous iteration times.

Demonstrated benchmarks show the system effortlessly handling grids up to 150^2, effectively outperforming parallel CPU cores by almost five-fold and single threads by $\approx 40\times$. Furthermore, the integrated '.h5' backed Qt GUI serves a robust pipeline for iterative testing and on-the-fly analytical visualization of distributions.

Future work will expand horizontally along several vectors including:

- Extension to true three-dimensional domains.
- Multi-GPU scaling via deep domain decomposition.
- Automated phase-diagram extraction algorithms mapping extensive structural regions seamlessly across the interactive GUI.

Acknowledgment

This work was performed as part of the Architecture of Computer Systems course at Ukrainian Catholic University. We thank the course instructors for guidance on parallelization strategies.

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